10.3 Rockets and Guns

WORKSHEET

The question

A railway truck, initially at rest, can move forwards without friction on a long straight horizontal track. On the truck, n guns are mounted parallel to the track and facing backwards, where $n > 1$. Each of the guns is loaded with a single projectile of mass m. The mass of the truck and guns (but not including the projectiles) is M.

When a gun is fired, the projectile leaves its muzzle horizontally with a speed $v - V$ relative to the ground, where V is the speed of the truck immediately before the gun is fired.

(i) All n guns are fired simultaneously. Find the speed, u, with which the truck moves, and show that the kinetic energy, K, which is gained by the system (truck, guns and projectiles) is given by

$$K = \tfrac{1}{2}nmv^2\left(1 + \frac{nm}{M}\right).$$

(ii) Instead, the guns are fired one at a time. Let u_r be the speed of the truck when r guns have been fired, so that $u_0 = 0$. Show that, for $1 \leq r \leq n$,

$$u_r - u_{r-1} = \frac{mv}{M + (n-r)m} \qquad (*)$$

and hence that $u_n < u$.

(iii) Let K_r be the total kinetic energy of the system when r guns have been fired (one at a time), so that $K_0 = 0$. Using $(*)$, show that, for $1 \leq r \leq n$,

$$K_r - K_{r-1} = \tfrac{1}{2}mv^2 + \tfrac{1}{2}mv(u_r - u_{r-1})$$

and hence show that

$$K_n = \tfrac{1}{2}nmv^2 + \tfrac{1}{2}mvu_n.$$

Deduce that $K_n < K$.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Before a gun fires, the truck moves at V. Afterwards the projectile's velocity is $V - v$ (that is, speed $v - V$ backwards). What is the projectile's velocity *relative to the truck before firing*?

STAGE 2 One at a time

Ask yourself Just before the rth gun fires, what is the mass of the truck with its remaining projectiles, and what is its speed? Write down conservation of momentum for the rth firing.

Pause and try Why does firing one at a time do worse here? When the first gun fires, it has to accelerate the truck *and* the $n-1$ unfired projectiles, so it produces less of a speed increase than it would on the bare truck.

STAGE 3 The energy bill

Ask yourself Write down the change in kinetic energy at the rth firing. How can $(*)$ help you simplify the term involving $u_r^2 - u_{r-1}^2$?

Try this next

1 **The other convention.** Redo parts (i) and (ii) supposing each projectile leaves at speed v relative to the truck *after* firing. Show that now firing one at a time gives the larger final speed.

2 **A comet in a dust cloud.** (From STEP III, 2009.) A comet of mass M enters a stationary dust cloud at speed V; after travelling a distance x its mass is $M(1+bx)$. If no other force acts, show that $(1+bx)v = V$ and find x in terms of t. If a constant force Mf acts in the direction of motion, use force = rate of change of momentum to show that $v = \frac{ft+V}{1+bx}$. Show that the comet can move at constant speed if $bV^2 = f$, and that otherwise its speed tends to $\sqrt{f/b}$.

3 **Two stages.** A rocket has exhaust speed v, empty mass M and fuel mass $9M$. Find its final speed. Now split it into two stages, each with empty mass $\frac{1}{2}M$ and fuel $4.5M$, the first stage being dropped when empty. Compare the final speeds.

4 **The inequality.** Prove that $\frac{x}{1+x} \le \ln(1+x) \le x$ for $x \ge 0$, by comparing $\frac{1}{1+s}$ with 1

and with $\frac{1}{1+x}$ on $[0, x]$.

Further reading

- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), Chapter 4. Momentum, variable mass and the rocket equation.
- **George Sutton & Oscar Biblarz, *Rocket Propulsion Elements*** (Wiley). The standard engineering text; the opening chapters are readable after this section.
- **Hermann Oberth, *Ways to Spaceflight*** (NASA translation, free online). A pioneer's own account.

The full guided solution, and what the question leads to at university, is in Section 10.3:
https://beyondstep.keithmcnulty.org/ch10/rockets.html